CMANint8

May 4, 2026

[1]:
```
import numpy as np
```

[2]:
```
W=np.array([[.85,-1.20,.34,2.10],[-0.07,0.91,-1.88,.12],[1.55,0.03,-0.44,-2.
↪31],[-0.18,1.03,0.77,0.55]])
```

[21]:
```
S=np.max(np.abs(W))/127
```

[23]:
```
S
```

[23]:
```
0.018188976377952755
```

[24]:
```
W_q=np.clip(np.rint(W/S).astype(int),-128,127)
```

[26]:
```
W_q
```

[26]:
```
array([[  47,  -66,   19,  115],
       [  -4,   50, -103,    7],
       [  85,    2,  -24, -127],
       [ -10,   57,   42,   30]])
```

[27]:
```
W_d=np.round(S*W_q,2)
```

[28]:
```
ABE=np.abs((W-W_d))
```

[30]:
```
ABE
```

[30]:
```
array([[0.  , 0.  , 0.01, 0.01],
       [0.  , 0.  , 0.01, 0.01],
       [0.  , 0.01, 0.  , 0.  ],
       [0.  , 0.01, 0.01, 0.  ]])
```

[31]:
```
MAE=np.mean(ABE)
MAE
```

[31]:
```
0.004375
```

[32]:
```
S_bad=.01
```

[33]:
```
W_star=np.clip(np.rint(W/S_bad).astype(int),-128,127)
W_star
```

[33]:
```
array([[  85, -120,   34,  127],
       [  -7,   91, -128,   12],
       [ 127,    3,  -44, -128],
       [ -18,  103,   77,   55]])
```

[34]:
```
W_dstar=np.round(S*W_star,2)
W_dstar
```

[34]:
```
array([[ 1.55, -2.18,  0.62,  2.31],
       [-0.13,  1.66, -2.33,  0.22],
       [ 2.31,  0.05, -0.8 , -2.33],
       [-0.33,  1.87,  1.4 ,  1.  ]])
```

[36]:
```
ABE=np.abs((W-W_dstar))
ABE
```

[36]:
```
array([[0.7 , 0.98, 0.28, 0.21],
       [0.06, 0.75, 0.45, 0.1 ],
       [0.76, 0.02, 0.36, 0.02],
       [0.15, 0.84, 0.63, 0.45]])
```

[37]:
```
MAE=np.mean(ABE)
MAE
```

[37]:
```
0.4225
```

The S should be a function of the max of the data and the range of integers being encoded. Otherwise, clipping will occur.